- **Label Noise:** Adding small random noise to the regression targets during training to improve robustness.

We conducted experiments using the Qwen2.5-7B model across all three scenarios: Long-Sequence (LongSeq), Reasoning, and Reinforcement Learning (RL). The results, measured in Mean Absolute Error (MAE), are presented in Table 6.

Table 6: Performance comparison (MAE ↓) of EGTP against alternative loss function designs on Qwen2.5-7B. Best results are bolded.

Method	LongSeq	Reasoning	RL
EGTP (Ours)	**83.65 ± 1.28**	139.14 ± 5.13	**92.78 ± 2.32**
Label Smoothing	158.44 ± 0.73	144.93 ± 19.60	268.88 ± 3.20
Log-scale Regression	135.73 ± 2.04	**136.09 ± 4.76**	208.02 ± 0.50
Label Noise	119.05 ± 2.25	217.44 ± 1.81	201.77 ± 1.91

The results show that while Log-scale regression is competitive on the Reasoning task, EGTP consistently outperforms others across all three domains, especially in LongSeq and RL, indicating better robustness.

D.3 Layer Selection Strategy

The practice of using final-layer representations for transfer learning, common in architectures like VGG and ResNet, persists in modern LLMs. Notably, systems such as Retrieval-Augmented Generation (RAG) (Li et al., 2025d) employ these activations for retrieval and classification, validating their semantic richness. Following this established paradigm, we rely on final-layer activations to perform length prediction.

To validate the rationale of using the final layer, we conducted experiments on the GSM8k dataset using hidden states from different layers of Qwen2.5-0.5B (which has 24 layers) to predict output length. As shown in Table 3, the performance of the final layer (Layer 24) is comparable to that of the best-performing intermediate layers (such as Layer 12 and Layer 16). Layer 24 achieves an RMSE of 92.95 ± 2.80, which is close to Layer 12's 93.47 ± 1.50. Although Layer 12 slightly outperforms in terms of MAE (70.26 ± 1.20), Layer 24's MAE (73.93 ± 2.10) still maintains a competitive level. Considering that using the final layer avoids over-engineering and unnecessary implementation complexity, we consistently utilize the hidden states from the last transformer layer across all model types and sizes in our final implementation.

Table 7: Performance of using different layers on Qwen2.5-0.5B (GSM8K). Metrics reported are RMSE and MAE (lower is better ↓).

Layer Index	RMSE ↓	MAE ↓
Layer 1	110.18 ± 2.50	85.91 ± 1.82
Layer 8	97.45 ± 1.81	81.29 ± 1.50
Layer 12	93.47 ± 1.50	**70.26 ± 1.20**
Layer 16	94.30 ± 1.64	70.79 ± 1.36
Layer 24 (Final)	**92.95 ± 2.80**	73.93 ± 2.10

E SUPPLEMENTAL EXPERIMENT RESULTS

E.1 BASELINES

We use the following methods as our baselines in the main experiment.

- **SSJF** (Qiu et al., 2024): Uses a fine-tuned BERT model, formulated either as a regression task to predict the absolute token length (**SSJF-Reg**) or as a multi-class classification task for length categories (**SSJF-MC**).
- **S3** (Jin et al., 2023): Utilizes a fine-tuned Distilbert model to classify the output into one of ten predefined length buckets.
- **PiA** (Zheng et al., 2023): Prompts or instruction-tunes a Vicuna model to predict its own response length.
- **TPV** (Eisenstadt et al., 2025): Trains a linear regressor to estimate the relative progress of DeepSeek-R1-Distill's reasoning process.
- **TRAIL** (Shahout et al., 2025): Trains an MLP classifier on the Llama's internal embeddings to predict the final output length.
- **LTR** (Fu et al., 2024): Employs an OPT model backbone, trained through classification (**LTR-C**) or ranking, to predict the output sequence length.

E.2 GENERALIZATION TO SUPER-LONG SEQUENCES

Most standard Supervised Fine-Tuning (SFT) datasets predominantly contain sequences shorter than 4k tokens (Bai et al., 2024b). Consequently, our main experiments focus on these typical distributions. However, to investigate the generalization capability of EGTP on Out-of-Distribution (OOD) samples with extreme lengths, we conduct additional experiments on the `euclaise/writingprompts` (Fan et al., 2018) dataset, where the longest sequence exceeds 17k tokens.

Table 8: Performance comparison on super-long sequences from the `euclaise/writingprompts` dataset ($>$ 17k tokens). EGTP demonstrates superior generalization capability in this OOD setting.

Metric	EGTP	SSJF-Reg	SSJF-MC	S3	PiA	TPV	TRAIL	LTR-C
MAE ↓	**195.89** ± 1.54	280.26 ± 6.63	315.25 ± 16.88	211.02 ± 27.95	214.48 ± 5.15	724.04 ± 16.80	212.73 ± 6.00	255.12 ± 20.46
RMSE ↓	**257.49** ± 43.18	366.32 ± 69.61	418.76 ± 19.52	281.78 ± 40.07	283.10 ± 13.74	1049.21 ± 143.57	279.42 ± 4.22	293.27 ± 11.18

The results in Table 8 demonstrate that EGTP maintains strong generalization even on sequences far exceeding the length distribution seen during training. It achieves the lowest MAE (195.89) and RMSE (257.49), significantly outperforming baselines such as TPV and SSJF-MC which fail to scale effectively. This suggests that the entropy-guided representations capture intrinsic autoregressive patterns related to generation termination, rather than merely memorizing length biases from the training set.

E.3 COMPUTATIONAL EFFICIENCY ANALYSIS

To offer a more granular quantitative analysis of the overhead introduced by different length prediction methods, we evaluated the inference latency and memory consumption of the prediction modules in isolation. This complements the end-to-end system performance reported in Section 4.4.

We utilized the GSM8K math reasoning dataset for evaluation. All experiments were conducted on a single NVIDIA RTX 4090 (24GB) GPU under consistent environmental settings. We measured:

- **Avg Time (ms):** The average wall-clock time required for the predictor to process a query and generate a length estimate.
- **Avg VRAM (MB):** The additional GPU memory required to load and run the prediction module.

We compared EGTP against baselines. Note that for PiA, which relies on the LLM itself to generate token counts, we exclusively recorded the time consumed for generating the specific length-prediction tokens to ensure a fair comparison.

Table 9: Auxiliary model overhead for predicting **Qwen2.5-7B** response lengths. EGTP incurs negligible latency and memory costs compared to methods requiring external models.

Metric	EGTP (Ours)	SSJF-MC	SSJF-Reg	S3	LTR-C	TPV	PiA	TRAIL
Avg Time (ms)	**0.67 ± 0.07**	3.94 ± 0.01	4.04 ± 0.12	2.26 ± 0.08	12.57 ± 0.07	0.83 ± 0.21	60.90 ± 0.33	2.04 ± 0.08
Avg VRAM (MB)	**7.21**	269.76	269.76	264.19	238.41	6.38	-	268.03

Table 10: Auxiliary model overhead for predicting **Llama3.2-3B** response lengths.

Metric	EGTP (Ours)	SSJF-MC	SSJF-Reg	S3	LTR-C	TPV	PiA	TRAIL
Avg Time (ms)	**0.65 ± 0.01**	3.95 ± 0.02	4.18 ± 0.13	2.41 ± 0.04	12.56 ± 0.03	0.79 ± 0.11	60.40 ± 1.97	2.59 ± 0.11
Avg VRAM (MB)	**5.52**	269.76	269.76	264.19	238.41	5.95	-	268.03

Tables 9 and 10 unequivocally demonstrate the efficiency of our approach. EGTP achieves the lowest inference time ($\approx$ 0.66ms), which is effectively negligible in the context of LLM inference. In contrast, auxiliary model-based methods like SSJF and TRAIL typically require 2–12ms to perform a forward pass on their external models. PiA is the slowest ($\approx$ 60ms) due to the overhead of autoregressive decoding for the prediction tokens.

Since EGTP reuses the hidden states already computed during the prefill phase of the main LLM, it only requires storing a lightweight linear head, consuming merely 5–7 MB of VRAM. Conversely, baselines requiring separate auxiliary models consume significantly more memory (238–270 MB) to store model weights, which can compete for resources with the main serving system.

E.4 Analysis of Prediction Stability

E.5 Length Prediction Performance on other datasets

Table 11 presents a comprehensive comparison of our proposed method, EGTP, against several baseline approaches using the Qwen2.5 model family (0.5B and 1.5B). The evaluation, based on Mean Absolute Error (MAE), is conducted across three distinct scenarios: Long Sequence, Reasoning, and RL. The results clearly demonstrate the superiority of our method. EGTP consistently achieves the lowest average MAE for both model sizes, significantly outperforming all competitors. The TRAIL method consistently secures the second-best position, but still trails our approach by a notable margin.

E.6 Detailed Prediction Accuracy Results

The following tables provide a detailed breakdown of the output length prediction results, supplementing the averaged scores presented in the main text. We evaluate several baseline methods on their ability to predict the response length from a given prompt. The prompts are sourced from LongBench, ZeroSCROLLS (See Table 12 and Table 13), and IFEval (See Table 14 and Table 15). The corresponding responses are generated by the LLMs specified in the "Model" column (Qwen2.5, Llama3.2, and DeepSeek-R1-Distill models).

Benchmark		Prediction Method							
Model	**Scenario**	**EGTP (Ours)**	SSJF-Reg	SSJF-MC	S3	PiA	TPV	TRAIL	LTR-C
Qwen2.5 0.5B	Long Seq	133.10	375.67	675.28	380.45	444.17	847.42	170.74	215.44
	Reasoning	145.85	221.15	466.00	200.91	296.02	597.00	146.24	178.28
	RL	88.52	125.40	224.68	179.19	255.52	261.36	161.79	199.14
	Avg.	**122.49**	240.74	455.32	253.52	331.90	568.59	**159.59**	197.62
Qwen2.5 1.5B	Long Seq	125.42	301.68	690.73	212.73	351.20	587.64	151.86	162.54
	Reasoning	146.48	310.64	815.78	169.72	432.05	584.08	138.01	139.02
	RL	96.59	129.65	173.95	167.97	133.82	208.39	160.60	170.93
	Avg.	**122.83**	247.32	560.15	183.47	305.69	460.04	**150.16**	157.50

Table 11: Comparison of different length prediction methods based on Mean Absolute Error (MAE). Lower values indicate better performance. In each 'Avg.' row, which shows the mean performance, the **best-performing** method is highlighted in red, and the **second-best** is highlighted in green.

Table 12: Performance of length prediction methods on Long-Sequence Scenario.

Model	Method	LongBench	ZeroSCROLLS
		MAE	MAE
Qwen2.5 -0.5B -Instruct	SSJF-Reg	374.14	140.36
	SSJF-MC	696.02	654.55
	S3	599.09	161.82
	PiA	325.00	146.50
	TPV	1078.33	616.52
	TRAIL	**176.19**	165.29
	LTR-C	339.78	**91.09**
	EGTP(Ours)	**114.42**	**95.44**
Qwen2.5 -1.5B -Instruct	SSJF-Reg	250.71	**92.07**
	SSJF-MC	720.45	660.99
	S3	300.91	124.55
	PiA	214.32	151.48
	TPV	705.29	469.98
	TRAIL	**147.95**	155.77
	LTR-C	207.55	117.53
	EGTP(Ours)	**93.36**	**82.74**

Table 13: Performance of length prediction methods on Long-Sequence Scenario (MAE only).

Model	Method	LongBench	ZeroSCROLLS
		MAE	MAE
	SSJF-Reg	208.55	**96.75**

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Qwen2.5 -3B -Instruct			